

TONGDE, China

WMO: 529570

Lat: 35.267N

Lon: 100.650E

Elev: 3290

StdP: 67.56

Time Zone: 8.00 (E08)

Period: 90-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.7	-23.8	-37.3	0.2	-15.9	-34.7	0.2	-15.6	13.9	-1.5	10.3	-2.3	3.8	90	0.737

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	21.9	11.1	20.3	10.4	18.9	9.7	13.4	18.8	12.4	17.4	11.5	16.1	2.7	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
11.6	12.8	15.4	10.6	12.0	14.3	9.7	11.3	13.6	49.7	19.0	46.5	17.2	43.5	16.3	16.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.0	8.0	6.5	-29.6	24.7	2.6	1.1	-31.5	25.5	-33.0	26.1	-34.5	26.8	-36.4	27.5
			-30.0	15.1	2.5	0.9	-31.8	15.7	-33.3	16.3	-34.7	16.8	-36.5	17.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.3	-12.0	-8.1	-2.8	2.8	7.4	10.0	11.7	11.5	7.9	1.6	-5.2	-10.0
	DBStd	8.85	3.91	3.65	3.69	3.20	2.94	2.59	2.28	2.42	3.07	3.14	3.17	2.94
	HDD10.0	3362	683	506	396	217	92	31	9	11	79	262	455	621
	HDD18.3	6227	941	739	654	466	340	250	207	211	314	520	705	879
	CDD10.0	177	0	0	0	1	11	31	61	58	15	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	10	0	0	0	1	0	1	1	6	2	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	3.1	3.2	3.3	3.7	3.5	3.4	2.9	2.6	2.5	2.6	2.7	3.1	3.2
Precipitation	PrecAvg	427	2	3	7	18	58	81	96	76	59	24	3	1
	PrecMax	599	7	13	26	58	123	138	152	143	112	93	14	7
	PrecMin	317	0	0	0	1	9	25	51	37	20	0	0	0
	PrecStd	77	2	3	6	11	26	26	31	26	22	17	4	2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.1	8.6	12.9	19.2	21.2	21.6	23.2	24.1	22.8	15.6	10.7	7.0
		MCWB	-2.8	-1.5	0.9	4.4	7.2	9.5	12.0	12.2	12.8	5.5	-0.1	-1.8
	2%	DB	2.1	5.9	10.7	16.1	18.5	19.7	21.3	21.9	19.6	12.9	8.3	4.1
		MCWB	-4.9	-2.4	0.0	3.6	6.7	9.7	11.7	11.3	10.0	4.2	-1.2	-3.7
	5%	DB	-0.2	3.6	8.4	13.6	16.5	18.4	19.8	20.2	17.1	11.0	6.5	1.8
		MCWB	-6.3	-3.7	-0.6	2.2	6.1	9.0	11.1	10.8	7.7	2.5	-1.9	-5.0
	10%	DB	-2.4	1.2	6.1	11.3	14.6	16.9	18.0	18.4	14.8	9.3	4.5	-0.5
		MCWB	-7.6	-5.1	-1.6	1.4	5.3	8.3	10.9	10.2	7.0	1.8	-2.9	-6.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.8	-1.0	2.3	6.2	9.8	12.7	14.3	15.1	14.4	6.7	0.6	-2.0
		MCDB	4.6	7.4	9.8	16.2	16.7	18.6	19.4	21.7	20.9	14.0	8.9	7.0
	2%	WB	-4.6	-2.3	0.8	4.3	8.4	11.5	13.1	13.5	11.8	5.1	-0.7	-3.5
		MCDB	1.7	5.5	7.9	13.6	15.5	16.7	18.4	18.7	17.0	10.7	6.5	3.7
	5%	WB	-6.1	-3.6	-0.2	3.2	7.3	10.6	12.4	12.3	10.0	3.9	-1.6	-5.0
		MCDB	-0.5	3.0	6.7	11.6	13.6	15.5	17.3	17.2	14.4	8.8	5.6	1.5
	10%	WB	-7.6	-5.0	-1.3	2.0	6.5	9.6	11.5	11.3	8.6	2.9	-2.7	-6.5
		MCDB	-2.7	0.8	5.2	9.9	12.3	14.5	16.0	15.7	12.6	7.3	4.2	-0.8
Mean Daily Temperature Range	5% DB	MDBR	19.3	17.9	16.0	14.5	12.4	11.3	11.2	12.9	13.4	14.5	19.5	20.5
		MCDBR	22.7	21.4	20.1	19.8	16.8	14.9	16.2	17.1	18.5	17.7	21.8	23.4
		MCWBR	16.7	14.1	11.7	9.5	7.5	6.4	7.4	7.5	8.6	9.4	13.3	16.6
	5% WB	MCDBR	21.6	19.8	16.9	16.4	13.0	11.4	11.8	11.9	12.5	14.1	20.2	22.6
		MCWBR	16.0	13.4	9.8	8.2	6.1	5.4	5.8	5.6	6.3	7.8	12.6	16.2
Clear-Sky Solar Irradiance	taub	0.218	0.252	0.303	0.341	0.346	0.337	0.325	0.319	0.296	0.267	0.229	0.208	
	taud	2.542	2.445	2.275	2.186	2.274	2.399	2.485	2.496	2.538	2.549	2.568	2.593	
	Ebn at Noon	1022	1006	969	941	936	940	949	953	967	977	997	1019	
	Edh at Noon	76	96	125	145	135	119	108	105	95	85	74	68	
All-Sky Solar Radiation	RadAvg	3.34	4.09	5.22	6.02	5.99	5.74	5.86	5.58	4.62	4.20	3.64	3.19	
	RadStd	0.21	0.20	0.25	0.29	0.30	0.47	0.58	0.44	0.38	0.28	0.14	0.10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)		+0.36	N/A	N/A	N/A	N/A	N/A	N/A	-114	+35	N/A

Nomenclature: See separate page